

# LISA

Light-based In-field Spectral Analyser



**Foldscope made microscopy cheap enough for everyone. LISA does the same for water chemistry.**

The 24-hour master plan for the Ashoka Startup Challenge: the product, the chemistry, the code, the build, the pitch, who does what, and how to win.

**Team:** Yash • Raj • Stuti • Aditi • Parth

**Technical foundation:** Feng et al., *Sensors & Actuators B* 462 (2026) 140011

**Included in this pack:** this plan (PDF) + a working web app ([lisa-app.html](#))

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## The three golden rules (read these out loud at kickoff)

1. **One test that really works beats six that sort of work.** Phosphate is the one. Iron is a bonus only if phosphate is done by Gate 2.
2. **Never fake a number.** The app shows other tests as "in development," which is true. If a judge catches one made-up result, we lose all our credibility. Being open about limits is how we win the science questions.
3. **Proof beats promises.** Our winning moment is a **blind sample chosen by a judge**, measured live, and matched against a lab spectrophotometer reading. Build everything toward that moment.

# 1 The 60-second version

What we're building, in one page. Everyone reads this before doing anything else.

## One-line pitch

LISA turns any smartphone into a pocket water lab for about ₹2,000. It uses a folding paper-style spectrometer, a tested colour reaction, and an app that gives the concentration, a safe or unsafe verdict, and a data point on a community water map.

## What we build in 24 hours

- **Hardware:** a black, Foldscope-style card body. Inside: a white LED, a 10 mm cuvette, a razor-blade slit, and a diffraction grating taped over the phone camera. Held on with a phone case or clip.
- **Chemistry: phosphate** by the molybdenum-blue method. It's the same chemistry the 2026 reference paper validated, so we're building on published work. **Iron** (phenanthroline) is test #2 if time allows.
- **App (already written, in this pack):** select test → blank → scan → spectrum → concentration → verdict against the limit → save, share and read the result aloud in Hindi.
- **Proof:** a calibration curve ( $R^2 \geq 0.99$ ), a limit of detection, a comparison against Raj's lab UV-Vis spectrophotometer, and a live blind test.

## What we pitch

- **Problem:** 2.1 billion people lack safely managed drinking water [8]. In India's 2024 groundwater survey, 19.8% of samples exceeded the nitrate limit, 13.2% iron and 9.04% fluoride [7]. **Sonipat is on the fluoride list.**
- **Gap:** labs are accurate but slow and far away. The field kits used by 24.8 lakh trained women under Jal Jeevan Mission [9] rely on matching colours by eye, which is subjective and leaves no data trail [10].
- **Solution:** the same kind of colour chemistry, read objectively by a phone spectrometer. Each new test is a new chemistry pack on the same device.
- **Proof:** our live numbers, plus the paper's result: under 10% error against a lab instrument in under 5 minutes [1].

**₹2–3k**

prototype cost, phone excluded  
(paper's device: under \$700 [1])

**< 10 min**

sample to result  
(lab: days)

**1 → many**

tests per device:  
swap the chemistry pack

## Who owns what (details in Section 12)

| Person | Owns                                                                  | Main deliverable by Gate 2 (T+8h)                                              |
|--------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Yash   | App, optics integration, technical demo                               | App deployed on the demo phone, rainbow captured, wavelengths calibrated       |
| Raj    | Chemistry, lab access, lab spectrophotometer reference, safety        | Reagents and 7 standards made; reference readings on the lab spectrophotometer |
| Stuti  | Pitch deck, story, research and citations, lead presenter             | Slides 1–6 drafted; problem and market numbers verified                        |
| Aditi  | Data logging, sample collection, blind-test coding, Q&A drills        | Filled data sheet; real samples collected and labelled                         |
| Parth  | Enclosure build from the template, procurement, video, demo logistics | Enclosure v1 built; buying list done; B-roll filming started                   |

## 2 How we win: what the judges will score

Most startup-challenge panels score some version of the five criteria below. Plan the 24 hours so each one gets a clear, visible answer.

| Criterion                         | What judges want                       | Our answer (make it visible)                                                                                                                                |
|-----------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Problem &amp; impact</b>       | A real, large, local pain with numbers | CGWB 2024 contamination stats, with Sonipat named for fluoride; JJM's 24.8 lakh women already testing by eye; a glass of hostel tap water on the table      |
| <b>Innovation</b>                 | Not "another app": a new approach      | "Foldscope for chemistry": a frugal phone spectrometer, modular chemistry packs, a sample-blank correction, and a community data layer                      |
| <b>Working prototype</b>          | It actually works in front of them     | A live blind test chosen by a judge; the calibration curve and lab-instrument comparison on a slide                                                         |
| <b>Feasibility &amp; business</b> | Who pays, how it scales, what it costs | Razor-and-blade model (device + reagent refills); government channel through JJM field kits; unit cost table                                                |
| <b>Team</b>                       | Can this team execute?                 | A mix of chemistry lab access (Raj), full-stack and hardware skills (Yash), and research and design (Stuti), shown by the fact that we built it in 24 hours |

### Our "unfair advantages." Use all four.

1. **A peer-reviewed 2026 paper** [1] already shows phone-spectrometer phosphorus testing at lab-grade accuracy. We aren't asking judges to believe the physics. We're showing a frugal, India-first version of proven science.
2. **A lab spectrophotometer head-to-head.** Raj runs the same standards on the chemistry department's UV-Vis. "Our ₹2,000 kit vs. the ₹X-lakh lab instrument" is one slide that changes the room.
3. **A judge-picked blind sample.** Theatre plus proof. Nobody else will do this.
4. **A plug-in to an existing government workflow.** JJM already trains women to run colour-chart field tests [9]. LISA reads those colours objectively and keeps the data. Adoption doesn't need a new behaviour.

### What loses hackathons like this

- Claiming it "detects any chemical" or "detects lead at safe levels." A chemistry judge will stop the pitch right there.
- A demo that depends on one fragile alignment with no backup. Section 14 gives four backup levels.
- No numbers on the results slide. Show  $R^2$ , limit of detection, and % error.
- Running out of time on slides. Freeze the deck at T+18h and rehearse three times.

## 3 The science, in plain words

Everyone on the team should be able to explain this page in 30 seconds.

### 3.1 From colour to concentration

Most dissolved contaminants are invisible. A **reagent** reacts with one specific substance, the **analyte**, to make a coloured compound. More analyte means deeper colour. For phosphate the colour is blue (molybdenum blue). For iron it's orange-red (the iron-phenanthroline complex).

A spectrometer shines white light through the coloured liquid and measures how much of each wavelength gets through. The **Beer-Lambert law** links the two:

$$A(\lambda) = -\log_{10}(I_{\text{sample}}(\lambda) / I_{\text{blank}}(\lambda)) = \epsilon(\lambda) \cdot \ell \cdot c$$

A = absorbance • I = light intensity •  $\epsilon$  = how strongly the compound absorbs •  $\ell$  = path length (10 mm cuvette) • c = concentration

Because A is proportional to c, we measure a few **standards** (known concentrations), fit a straight line, and read unknown samples off that line. That's the **calibration curve**. It's the heart of the demo.

### 3.2 Why a phone can be a spectrometer

A phone already has three of the four parts of a spectrometer: a light source (the white LED flash), a detector (the camera sensor) and a computer. The missing part is something that splits light into its colours. A cheap **diffraction grating** does that, turning the slit image into a rainbow across the camera frame. Each pixel column then corresponds to a wavelength. We find the pixel-to-wavelength mapping by photographing a CFL tube light, whose mercury lines at 436, 546 and 611 nm act as free rulers [11,13].

### 3.3 What the reference paper proved (Feng et al., 2026 [1])

| Result in the paper                           | Number                                                                | How we use it                                                   |
|-----------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------|
| Device size / weight                          | 69×47×26 mm, 35 g                                                     | A target for our v2 form factor                                 |
| Range, spectral interval, wavelength accuracy | 400–700 nm • 0.78 nm • ±1 nm                                          | Our design range is the same 400–700 nm                         |
| Signal-to-noise; intensity variation          | >500:1; <1.5%                                                         | The benchmark we compare our own numbers against                |
| Total phosphorus accuracy                     | Relative error <10%, RSD <1%                                          | Proof that the chemistry plus phone approach works              |
| Time per test                                 | Over 40 min (lab) → under 5 min                                       | Headline speed claim (cite it as theirs)                        |
| Real wastewater, 20 samples                   | MAE 0.020 mg/L; 16/20 within 10%; paired t-test p = 0.185             | Honest framing: good for field screening, not a lab replacement |
| AI model (GAF + 2D-CNN), 480 spectra          | R <sup>2</sup> 0.996, RMSE 0.0076 mg/L (vs PLSR R <sup>2</sup> 0.960) | Our v2 model once we have that much data                        |
| Price                                         | Under \$700 (fibre spectrometer >\$4,000; benchtop >\$10,000)         | We go further: about ₹2–3k prototype, Foldscope-style           |

|                                |                                            |                                                                    |
|--------------------------------|--------------------------------------------|--------------------------------------------------------------------|
| Cross-phone calibration matrix | $R^2$ improved from 0.42–0.64 to 0.91–0.97 | Our per-phone calibration step, and the roadmap for a shared model |
|--------------------------------|--------------------------------------------|--------------------------------------------------------------------|

### Say this precisely

The paper measured *total phosphorus* using a commercial kit and only filtered the samples, without digestion. Their own discussion says the portable method measures orthophosphate-type forms, so real-sample agreement is a "practical consistency check," not strict equivalence. We measure **orthophosphate**, and we say so. Precise wording like this earns trust from technical judges.

## 3.4 How LISA is different from the paper

| Feng et al. 2026                                                 | LISA                                                                                            |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Machined optics, quartz cuvette, fibre coupling, 3D-printed body | Folding card body, plastic cuvette, film grating, razor-blade slit (the Foldscope approach [2]) |
| Under \$700 retail                                               | About ₹2–3k prototype, with a target of a few hundred rupees in hardware at scale (estimate)    |
| One analyte (TP), China, lab/industry users                      | Modular chemistry packs chosen for India: phosphate, iron, fluoride (Sonipat), nitrate          |
| Cloud model, English UI                                          | Works offline, Hindi voice result, WhatsApp share, community map, field-kit (FTK) integration   |

## 4 Which test we build (and why not lead)

Decision: **Phosphate first, iron second**. Lead goes on the roadmap slide as a research track.

| Candidate | Why it matters                                                                     | Chemistry ( $\lambda$ )                                                      | 24h risk                                                                                                   | Verdict                                                        |
|-----------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Phosphate | Marker of sewage, detergent and fertiliser runoff; the paper's validated chemistry | Molybdenum blue, read 630–690 nm on a phone (EPA 365.3 reads at 650 nm [4])  | <b>Low</b> Common lab reagents; strong blue colour                                                         | <b>Test #1</b>                                                 |
| Iron      | IS 10500 parameter [6]; 13.2% of CGWB samples exceed it [7]; rusty hand-pump water | 1,10-phenanthroline, ~510 nm, the centre of the camera's best range          | <b>Low</b> Standard undergraduate experiment                                                               | <b>Test #2</b> (stretch)                                       |
| Fluoride  | Sonipat flagged in CGWB 2024 [7]; causes fluorosis                                 | SPADNS–zirconium, ~570 nm; colour <i>bleaches</i> as fluoride rises          | <b>Medium</b> Reagent may not be on the shelf; interferences                                               | Roadmap: "next pack." Try only if SPADNS is available          |
| Nitrate   | 19.8% of samples exceed the limit [7]; blue-baby syndrome                          | Cadmium/zinc reduction + Griess, ~540 nm                                     | <b>Medium</b> Reduction step is fiddly                                                                     | Roadmap                                                        |
| Lead      | Highly toxic; limit 0.01 mg/L = 10 $\mu$ g/L [6]                                   | Dithizone needs chloroform extraction; rhodizonate is only semi-quantitative | <b>High</b> Simple colour tests can't reliably reach 10 $\mu$ g/L; even commercial home kits struggle [18] | Research track: pre-concentration or an electrochemical add-on |

### How to answer "Why phosphate if it isn't in the drinking-water standard?"

"Phosphate is a **contamination tracer**. Clean groundwater has very little, so a high reading means sewage, detergent or fertiliser is getting in. That usually comes with pathogens and nitrate. We chose it because it's the chemistry the 2026 paper validated, so our first pack stands on published science. The platform is the point: iron is our second pack, and fluoride, Sonipat's own problem, is next."

### Reference levels the app uses

| Test               | "Caution" level | "Unsafe/Alert" level | Source                                                                                            |
|--------------------|-----------------|----------------------|---------------------------------------------------------------------------------------------------|
| Phosphate (as P)   | 0.05 mg/L       | 0.1 mg/L             | US EPA "Gold Book" stream guidance [21]. Not a BIS drinking-water parameter; used as an indicator |
| Iron               | 0.3 mg/L        | 1.0 mg/L             | IS 10500:2012 [6] (the 0.3 mg/L acceptable limit in the original text)                            |
| Fluoride (roadmap) | 1.0 mg/L        | 1.5 mg/L             | IS 10500:2012 [6]                                                                                 |
| Nitrate (roadmap)  | —               | 45 mg/L              | IS 10500:2012 [6]                                                                                 |
| Lead (research)    | —               | 0.01 mg/L            | IS 10500:2012 [6]                                                                                 |

Raj: check the current IS 10500 amendment for iron on the BIS PDF before finalising the slide. The app's limits can be edited in its code (ANALYTES list).

## 5 How LISA works: the whole system

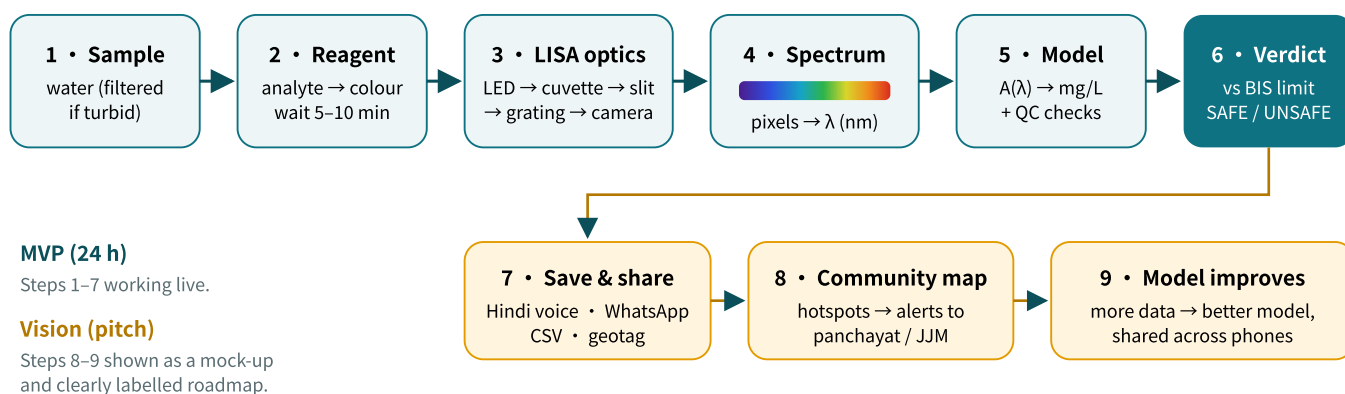


Figure 1. The end-to-end pipeline. Blocks 1–7 are the MVP. Blocks 8–9 are the platform vision for the pitch.

### 5.1 Optical path (side view)

t the tube  $\approx 33^\circ$  from the phone's normal (1000 lines/mm) so the green part of the first-order spectrum lands mid-frame

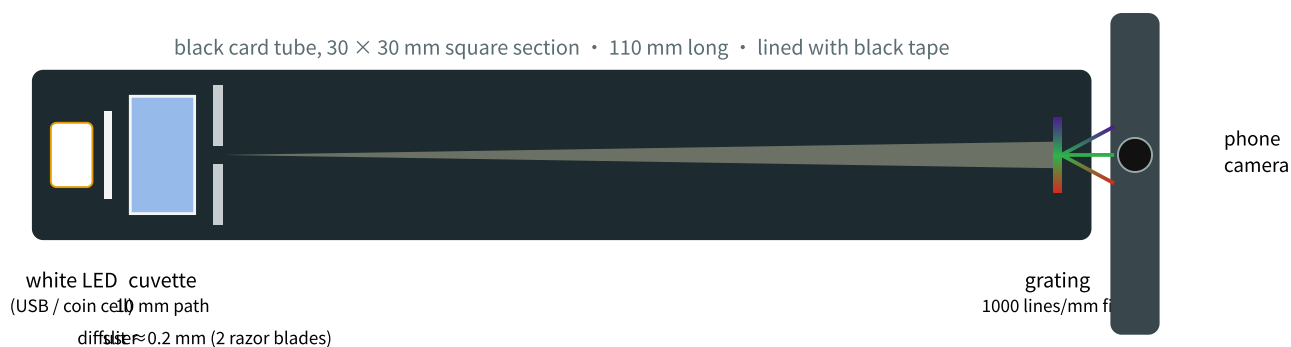


Figure 2. Plan-A layout (drawing not to scale; the template in Appendix B is 1:1). v1 uses its own LED because it's easier to align. The paper's trick of borrowing the phone's own flash through a fibre is the v2 upgrade.

#### Grating angle rule of thumb

$\sin \alpha \approx \lambda_{\text{centre}} / d$ , where  $d$  is the line spacing. For 1000 lines/mm ( $d = 1000$  nm) and  $\lambda = 550$  nm,  $\alpha \approx 33^\circ$ . For a DVD-R piece ( $d \approx 740$  nm),  $\alpha \approx 48^\circ$ . For 500 lines/mm,  $\alpha \approx 16^\circ$ . **In practice:** tape the grating on the camera, hold the tube by hand, rotate until the rainbow sits in the middle of the screen, *then* fix that angle with the card wedge.



## 6 Hardware build guide + bill of materials

Owners: **Parth** (body and mount, working from the template) and **Yash** (optics and alignment). Raj gets the grating and cuvettes from the physics and chemistry labs.

### 6.1 Bill of materials

| Item                                                | Spec / notes                                                                                                 | Where (fastest first)                                                          | Approx. ₹           |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------|
| Diffraction grating                                 | Transmission film, 500 or 1000 lines/mm.<br>Backup: a piece of DVD-R with the layers peeled apart            | Physics teaching lab (Raj) • a blank DVD-R from anyone • online (not same-day) | 0–500               |
| Cuvettes ×6                                         | 10 mm path, plastic (PS/PMMA fine for visible light) or glass                                                | Chemistry lab (Raj)                                                            | 0–300               |
| White LED light                                     | USB LED stick (runs off a power bank = stable) or 5 mm white LED + CR2032 + switch                           | Stationery / electronics shop, quick-commerce app                              | 50–150              |
| Power bank                                          | Any; keeps LED brightness steady                                                                             | Team                                                                           | 0                   |
| Black card                                          | 300 gsm A4 ×4 (or any card + black paint/marker)                                                             | Stationery                                                                     | 40–80               |
| Matte black tape                                    | Electrical/insulation tape, 2 rolls. Lines the tube, kills stray light                                       | Stationery / hardware                                                          | 40–80               |
| Razor blades ×2                                     | Double-edge blades for the slit. Tape the sharp edges and handle carefully                                   | Pharmacy / general store                                                       | 10–20               |
| Diffuser                                            | Tracing paper or white plastic from a milk carton                                                            | Anywhere                                                                       | 0–10                |
| Phone mount                                         | Cheap back cover for the demo phone <i>or</i> a mobile clip. Glue the body to the cover so alignment repeats | Mobile accessories shop                                                        | 100–200             |
| Magnets (optional)                                  | 4 small neodymium discs, for detachable mounting like the paper [1]                                          | Hardware / online                                                              | 100–150             |
| Glue gun + sticks, cutter, steel ruler, cutting mat | —                                                                                                            | Makerspace / hostel                                                            | 0–200               |
| Reagents & glassware                                | See Section 7; phosphate chemicals cost a few rupees per test                                                | Chemistry lab (Raj)                                                            | ~0–500              |
| Syringe filters 0.45 µm (optional)                  | For turbid real samples, as in the paper [1].<br>Backup: filter paper + funnel                               | Chemistry / biology lab                                                        | 0–300               |
| <b>Prototype total (phone excluded)</b>             |                                                                                                              |                                                                                | <b>≈ ₹500–2,500</b> |

Prices are rough estimates for planning. Parth: update this table with real receipts for the business slide ("we built it for ₹\_\_\_").

### 6.2 Build steps (about 2.5 hours)

1. **Print Appendix B at 100% scale** (turn off "fit to page"). Check the 50 mm scale bar with a ruler.

2. Glue the print onto black card (or cut plain card and blacken the inside). Cut on the solid lines, score and fold on the dashed lines.
3. **Main tube (A):** fold into a 30×30 mm square tube and glue the tab. Line the inside with matte black tape. *Stray light is the enemy.*
4. **Cuvette chamber (B):** cut the two opposite windows, fold, glue. A cuvette should slide in snugly from the top. Glue it across the far end of the tube.
5. **Slit plate (C):** tape two razor blades edge-to-edge over the window, leaving a gap about the thickness of one sheet of paper (~0.1–0.2 mm). Narrower gives sharper lines but less light. Put the slit between the cuvette chamber and the tube, with the slit running across the dispersion direction.
6. **Light end (D):** LED behind the diffuser, on the outer side of the cuvette chamber. Tape it light-tight. Mark "LED side" and "tube side" on the chamber.
7. **Grating:** tape a 15×15 mm square of grating over the phone's *main* rear camera. Line direction matters: the rainbow must spread across the frame, not along the slit. If it doesn't, rotate the film 90°.
8. **Find the angle:** open LISA → Setup → Start camera. Point the open end of the tube at the grating and rotate until the full rainbow (blue to red) sits in the frame. Note the angle.
9. **Wedge (E):** cut the triangle nearest that angle, glue the tube onto the wedge and the wedge onto the phone case. Cover every gap with black tape.
10. **Shroud:** put a black card hood around the phone-to-tube junction so no room light reaches the camera. Test: LED off should give a black frame.
11. **Lock it:** once the rainbow is good, don't touch the grating again. If it moves, redo the wavelength calibration (5 min).
12. Build **two** copies if time allows (Parth builds v2 while v1 is being tested). A spare saves the demo.

## 6.3 Troubleshooting

| Symptom                                       | Likely cause                                        | Fix                                                                                                       |
|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| No rainbow, only a white line                 | Looking at the zero order                           | Increase the tilt angle; check the grating isn't upside-down or rotated                                   |
| Rainbow cut off at one end                    | Angle slightly off / field of view                  | Adjust the wedge by a few degrees; move the slit further away; use 500 lines/mm                           |
| Blurry rainbow, no sharp CFL lines            | Slit too wide / camera not focused                  | Narrow the slit; tap to focus on the slit; lengthen the tube (the phone can't focus closer than ~7–10 cm) |
| Live plot says SATURATED                      | Too much light                                      | Lower exposure (slider); move the LED back; add a diffuser layer                                          |
| Blank changes between shots                   | Auto-exposure / white balance drifting; LED dimming | Press "Lock exposure"; power the LED from USB; always blank right before the sample                       |
| Red end noisy (>670 nm)                       | Phone IR-cut filter + weak LED red                  | Normal (the paper sees the same [1]). The phosphate band is 630–690 nm. Average 30 frames                 |
| Readings jump when the cuvette is re-inserted | Cuvette not seated the same way                     | Mark one side of the cuvette; always insert it facing the same way; wipe it with tissue                   |

### iPhone users

Mobile Safari won't let web apps lock exposure. On iPhone, use **Photo upload** mode: in the Camera app, long-press on the rainbow for "AE/AF LOCK," take the photo, and load it into LISA. **The demo phone should be an Android on Chrome**, which supports exposure lock and torch control.

## 7 Chemistry procedures

Owner: **Raj** (the only person who handles concentrated acid). Aditi assists with labelling, logging and safe dilutions under Raj's supervision.

### Safety first (non-negotiable)

Lab coat, goggles and nitrile gloves. Always add **concentrated  $\text{H}_2\text{SO}_4$  to water**, never water to acid, slowly and in a fume hood. Ammonium molybdate, antimony potassium tartrate and 1,10-phenanthroline are harmful or toxic: no skin contact, and dispose of waste in the lab's labelled containers. Nobody without lab training handles stock reagents. Demo samples on the pitch table must be **pre-reacted, capped cuvettes**. No open acid at the venue.

### 7.1 Test #1: Phosphate (molybdenum blue, ascorbic acid, two-reagent)

Based on US EPA Method 365.3 [4] and Murphy & Riley [3], scaled down to 10 mL. How it works: orthophosphate + molybdate (acid, antimony catalyst) → phosphomolybdate → reduced by ascorbic acid → **molybdenum blue**. The peak is ~880 nm but the colour absorbs strongly across 600–700 nm, which the phone can see. EPA 365.3 itself reads at 650 nm.

| Reagent                | How to make (Raj)                                                                                        | Keeps               |
|------------------------|----------------------------------------------------------------------------------------------------------|---------------------|
| R1: 11 N sulfuric acid | Slowly add 31 mL conc. $\text{H}_2\text{SO}_4$ to ~60 mL water, cool, make up to 100 mL                  | Months              |
| R2: Molybdate–antimony | 0.8 g ammonium molybdate + 0.02 g antimony potassium tartrate, dissolve and make up to 100 mL with water | Weeks (dark bottle) |
| R3: Ascorbic acid      | 6 g ascorbic acid in water, make up to 100 mL. <b>Make fresh</b> (fridge, a few days)                    | Days                |
| Stock P, 100 mg P/L    | 0.4393 g dried $\text{KH}_2\text{PO}_4$ in 1 L water                                                     | Months (fridge)     |
| Working P, 1.0 mg P/L  | 1.0 mL stock → 100 mL water                                                                              | Make daily          |

### Standards (10 mL each, made from the 1.0 mg P/L working solution)

| Std             | 0         | 1    | 2    | 3    | 4    | 5    | 6    |
|-----------------|-----------|------|------|------|------|------|------|
| mg P/L          | 0 (blank) | 0.10 | 0.20 | 0.40 | 0.60 | 0.80 | 1.00 |
| mL working soln | 0         | 1.0  | 2.0  | 4.0  | 6.0  | 8.0  | 10.0 |
| mL water        | 10.0      | 9.0  | 8.0  | 6.0  | 4.0  | 2.0  | 0    |

### Procedure (identical for standards, samples and blanks)

- 10.0 mL sample into a clean, labelled tube. (Filter first if turbid.)
- Add 0.2 mL R1 (acid), then 0.8 mL R2 (molybdate), and mix.
- Add 0.4 mL R3 (ascorbic), mix, **start a 5-minute timer**. Keep the timing the same for every tube.
- Pour ~3 mL into a cuvette, wipe it, insert it (marked side facing the same way every time), and scan.
- Stability check (do this overnight):** re-read standard 4 at 15 min, 1 h and 3 h. If A holds within  $\pm 2\%$ , demo samples can be pre-reacted before the pitch. If not, react them 10 minutes before going on stage.

Glassware: phosphate sticks to glass and many detergents contain phosphate. Rinse tubes with dilute acid and then distilled water, never with soap. Working range with a 10 mm cuvette is roughly 0.05–1.2 mg P/L. Above that, dilute 1:10 and set Dilution  $\times 10$  in the app. Unit conversion:  $\text{mg PO}_4/\text{L} = 3.07 \times \text{mg P/L}$ .

## 7.2 Test #2 (stretch): Iron (1,10-phenanthroline)

Standard method (APHA 3500-Fe B [5]; the classic teaching-lab experiment). Hydroxylamine reduces  $\text{Fe}^{3+}$  to  $\text{Fe}^{2+}$ . Three phenanthrolines bind  $\text{Fe}^{2+}$  to make an orange-red complex,  $\lambda_{\text{max}} \approx 508\text{--}510\text{ nm}$ ,  $\epsilon \approx 11,100\text{ L mol}^{-1}\text{ cm}^{-1}$ , stable between pH 2 and 9.

| Reagent              | Recipe                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------|
| Hydroxylamine • HCl  | 10 g / 100 mL water                                                                                   |
| 1,10-phenanthroline  | 0.1 g / 100 mL water (warm gently to dissolve)                                                        |
| Sodium acetate 1.2 M | 9.8 g anhydrous (or 16.3 g trihydrate) / 100 mL                                                       |
| Fe stock 10 mg/L     | 0.0702 g ferrous ammonium sulfate hexahydrate + 2.5 mL conc. $\text{H}_2\text{SO}_4$ , make up to 1 L |

**Per 10 mL sample/standard:** + 0.1 mL hydroxylamine, + 1.0 mL phenanthroline, + 0.8 mL acetate. Mix, wait 10 min, scan. **Standards:** 0, 0.25, 0.5, 1.0, 2.0, 3.0 mg Fe/L (0, 0.25, 0.5, 1, 2, 3 mL stock made up to 10 mL).

## 7.3 Real samples to collect (Aditi + Parth)

| Sample                                 | Why it's in the demo                                                          |
|----------------------------------------|-------------------------------------------------------------------------------|
| Distilled / RO water                   | Should read "below detection": shows no false alarms                          |
| Ashoka hostel tap and water cooler     | "Your water, tested today": a relatable number for the judges                 |
| Pond, canal or drain water (Sonipat)   | Real contamination; use a <b>sample blank</b> (see 8.4) because it's coloured |
| Tap water + a pinch of DAP fertiliser  | "Farm runoff": a vivid, controlled positive                                   |
| Laundry detergent rinse water          | Detergents can contain phosphates; a relatable story                          |
| Hand-pump or borewell water (for iron) | Rusty water story, and a real IS 10500 parameter                              |

Label each bottle with the sample ID, place, time and a photo. Store in the fridge. Test within 24 h.

## 8 Calibration and validation protocol

This page produces the numbers for the results slide. Owners: Yash + Raj run it, Aditi logs it.

### 8.1 Instrument calibration (once per build, 15 min)

1. **ROI:** LED on, blank cuvette in → Setup → Auto-detect ROI. Check that the cyan box covers the rainbow band.
2. **Exposure:** Lock exposure (Android). The live plot peak should be ~150–220 and never "SATURATED."
3. **Wavelength:** take out the LED end or open the chamber and point the slit at a white CFL tube light → Capture lamp spectrum → Assign CFL lines. The app fits  $\lambda = a \cdot px + b$ .  $R^2$  should be  $\geq 0.999$ . No CFL? Use a red (650 nm) and a green (532 nm) laser pointer and enter the two pairs manually.
4. **Dark frame (optional):** LED off → Capture dark frame.

### 8.2 Chemical calibration (per test, per phone, ~45 min)

1. Reagent blank → Calibrate → Capture blank.
2. Scan each standard from low to high, **3 replicates each** (re-insert the cuvette between replicates). That's 21 points for phosphate.
3. Fit calibration. The app reports  $R^2$ , LOD, LOQ, and the leave-one-out error for (a) single-band Beer–Lambert and (b) the full-spectrum "AI" ridge model, and uses whichever is better.
4. Export the calibration JSON as a backup (Setup → Backup).

### 8.3 Acceptance criteria (Gate 2)

| Metric                    | Target                                             | How it's computed                                                                        |
|---------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------|
| Linearity                 | $R^2 \geq 0.99$ (minimum 0.98)                     | Least squares $A = k \cdot C + b$ over all replicates                                    |
| Limit of detection (LOD)  | $\leq 0.1$ mg P/L (goal 0.05)                      | $LOD = 3.3 \cdot s_{y/x} / k$ ; $LOQ = 10 \cdot s_{y/x} / k$                             |
| Repeatability             | $RSD \leq 5\%$ at 0.4 mg/L                         | 5 re-insertions of the same cuvette                                                      |
| Accuracy                  | Recovery 90–110%                                   | 3 blind samples (Aditi codes them) + 1 spike recovery                                    |
| Agreement with lab UV-Vis | $ \text{error}  \leq 10\%$ (paper's threshold [1]) | Same cuvettes read on the chem-lab spectrophotometer at 650 nm (and 880 nm if available) |

### 8.4 Real-sample tricks (these make judges nod)

- **Sample blank:** for coloured or turbid water, use the same water *without* the molybdate reagent (replace it with water) as the blank. The app's "Blank type → Sample blank" option. This cancels the water's own colour, which is the paper's main source of real-sample error [1].
- **Spike recovery:** split a real sample. Add a known 0.2 mg P/L to one half.  $\text{Recovery} = (C_{\text{spiked}} - C_{\text{unspiked}}) / 0.2 \times 100\%$ . This proves the water's matrix isn't fooling us.
- **Dilution:** above range → 1:10 dilution → Dilution  $\times 10$ . The app multiplies the result back.

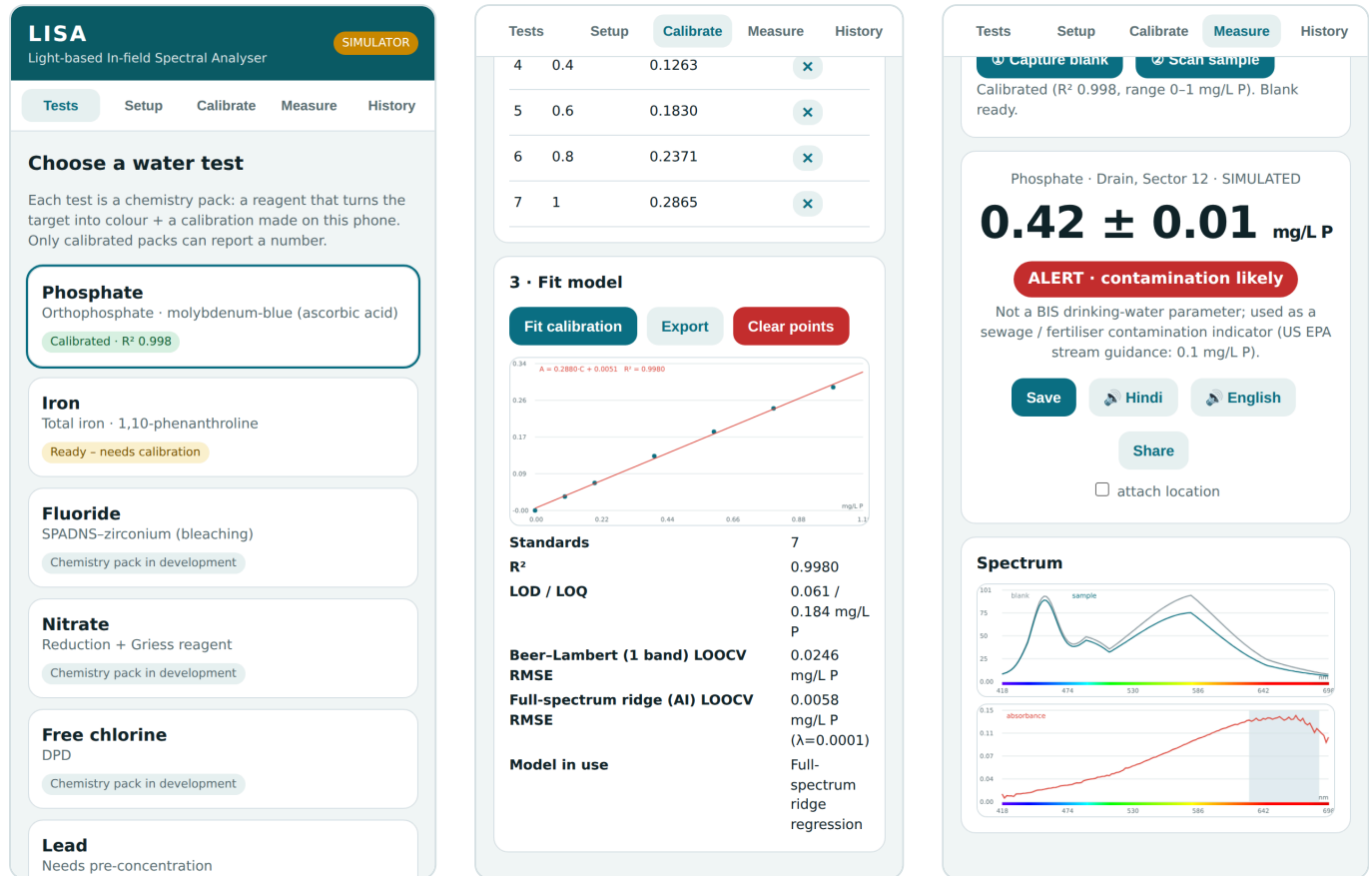
### 8.5 Data sheet (Aditi prints or keeps it in Google Sheets)

The app's History tab has a "Lab" column. Type the UV-Vis value there and % error appears automatically. Export CSV → paste into the deck chart.

| ID | Type<br>(std/blind/real/spike) | True or lab<br>value | LISA<br>value | %<br>error | A<br>(band) | Time<br>reacted | Phone /<br>build | Notes |
|----|--------------------------------|----------------------|---------------|------------|-------------|-----------------|------------------|-------|
|    |                                |                      |               |            |             |                 |                  |       |
|    |                                |                      |               |            |             |                 |                  |       |

## 9 The app: what's built, how it works, how to deploy

A complete, working web app ships with this plan: `lisa-app.html` (one file, no dependencies, works offline once loaded). It was tested end-to-end in its built-in simulator. Yash owns it: deploy, test on real hardware, polish.



Tests screen: live packs vs packs in development

Calibration curve with  $R^2$ , LOD, and linear vs AI error

Result, verdict, voice, share (simulator run shown)

### 9.1 The seven user steps and where they live

| Step                    | Screen              | What happens in code                                                                                                      |
|-------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1 Select analyte        | Tests               | Chemistry packs are config entries (ANALYTES): band, units, limits, standards, reagent text                               |
| 2 Blank calibration     | Calibrate / Measure | Averages 15–30 frames; column-wise mean of R, G, B inside the ROI → intensity profile (the paper's § 4.1 method [1])      |
| 3 Scan sample           | Measure             | Same capture; <b>cross-correlation alignment</b> to the blank ( $\pm 10$ px) corrects small shifts, as the paper does [1] |
| 4 Display spectrum      | Measure             | Pixels → $\lambda$ via the CFL fit; resampled 400–700 nm every 2 nm; $A(\lambda) = -\log_{10}(I_s/I_b)$                   |
| 5 Predict concentration | Measure             | Beer-Lambert band model or full-spectrum ridge model (whichever had lower cross-validated error), $\pm$ RMSE              |

|                      |         |                                                                                          |
|----------------------|---------|------------------------------------------------------------------------------------------|
| 6 Compare with limit | Measure | SAFE / CAUTION / UNSAFE pill; shows "< LOD" when below detection                         |
| 7 Save result        | History | Local storage, optional GPS, CSV export, WhatsApp/share, Hindi and English voice readout |

## 9.2 Built-in quality control (a strong talking point)

- Saturation warning (pixels  $\geq 250$ ) • blank too dark • spectrum shifted  $\geq 6$  px since the blank (cuvette re-seated badly)
- Above calibrated range  $\rightarrow$  tells you to dilute 1:10 • below LOD  $\rightarrow$  reports "< LOD" instead of a fake number
- Unexpected absorbance outside the analyte band  $\rightarrow$  "sample may be turbid or coloured: use a sample blank"
- Calibration mode check: a model made in RGB mode can't be used in spectrometer mode by mistake

## 9.3 Three input modes, so the demo never dies

| Mode                       | When                                                                                                                                        |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phone camera (live)</b> | Plan A. Android + Chrome. Supports exposure lock, exposure slider and torch                                                                 |
| <b>Photo upload</b>        | iPhone, or any phone where live camera access fails. Each capture opens the camera or gallery                                               |
| <b>Simulator</b>           | Rehearsal and training only. Physics-based synthetic spectra with noise. <b>Every result is stamped SIMULATED.</b> Never present it as real |

Also: **Spectrometer vs Colorimeter (RGB) mode.** If the grating fails, point the camera straight at the lit cuvette. The app then uses red/green/blue absorbance (phosphate  $\rightarrow$  red channel, iron  $\rightarrow$  green). Same workflow, lower resolution. This is backup Plan B.

## 9.4 Deploy in 10 minutes (camera needs HTTPS)

1. Rename `lisa-app.html`  $\rightarrow$  `index.html`.
2. **Netlify Drop** ([app.netlify.com/drop](https://app.netlify.com/drop)): drag the folder in and you get an https link. Or GitHub Pages: new repo  $\rightarrow$  upload `index.html`  $\rightarrow$  Settings  $\rightarrow$  Pages.
3. On the demo phone, open the link in Chrome  $\rightarrow$   $\vdots$   $\rightarrow$  "Add to Home screen." Now it opens like an app.
4. Make a QR code of the link for the last slide and the handout.

## 9.5 Yash's polish list (in priority order, stop when time runs out)

1. Test on the real rig; tune default frames (15–30) and the band limits if the real spectrum differs (edit band in ANALYTES).
2. Add the team logo and colours; set "Ashoka Startup Challenge demo build" in the header.
3. Add a 5-minute reaction timer button on the Measure screen.
4. Add a simple "Community map" page: plot saved results with lat/lon (Leaflet needs internet; a static mock-up in the deck is fine).
5. Optional cloud sync (Firebase/Supabase) for the map. **Not needed to win.** Show it as roadmap.

### Honest caveat on the screenshots

The screenshots above come from the simulator, where noise is idealised ( $\pm 0.01$  mg/L). Real hardware will be noisier. Put the **real** numbers from Section 8 on the results slide, not these.



## 10 The "AI": what to build and what to say

Judges love AI and distrust AI hype. Here is the version that's true and still impressive.

### What LISA's AI actually does (MVP)

1. **Learned calibration:** a full-spectrum ridge regression (a close cousin of PLS [19], which the paper uses as a baseline) learns from all wavelengths, not just one. The app compares it against plain Beer–Lambert with leave-one-out cross-validation and uses whichever is better.
2. **Automatic quality control:** it flags saturation, misalignment, out-of-range and matrix-interference problems a non-expert wouldn't spot.
3. **Computer-vision ROI + drift correction:** it finds the spectrum in the image and aligns it by cross-correlation.

### Roadmap AI (say "next")

1. Paper-style **GAF + 2D-CNN** model [1] once we have ~500 spectra per analyte (the paper used 480 and got  $R^2$  0.996).
2. **Cross-phone calibration matrix:** one shared model for any phone, from a reference card or standards [1].
3. **Kinetic reading:** record colour development over time to reject interferences. The paper lists this as future work.
4. **Anomaly maps:** spot contamination hotspots from community data.

### Never say

"AI detects any contaminant from a photo." The chemistry does the detecting. The AI makes the reading more accurate, more robust and easier for anyone to use.

### Optional 2-hour notebook for Yash or Raj (if the team wants a separate "AI" slide)

```
# Export History/calibration JSON from the app, then in Python:
import json, numpy as np
from sklearn.cross_decomposition import PLSRegression
from sklearn.linear_model import Ridge
from sklearn.model_selection import LeaveOneOut, cross_val_predict
cal = json.load(open('lisa-calibration-phosphate.json'))['cal']['points']
X = np.array([a if a is not None else 0 for a in p['A']] for p in cal); y = np.array([p['conc'] for p in cal])
for name, m in [('PLS-3', PLSRegression(3)), ('Ridge', Ridge(1e-3))]:
    pred = cross_val_predict(m, X, y, cv=LeaveOneOut()).ravel()
    print(name, 'RMSE', np.sqrt(np.mean((pred-y)**2)).round(4), 'R2', (1-np.sum((pred-y)**2)/np.sum((y-y.mean())**2)).round(4))
```

# 11 Upgrades that take it from 9/10 to 10/10

Ranked by value for time. Items 1–5 are part of the 24-hour plan. Items 6–10 go on slides as vision.

| #  | Upgrade                                                                                                           | Why it's powerful                                                                                            | Cost              |
|----|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------|
| 1  | <b>Judge's blind sample:</b> three coded vials, a judge picks one, LISA reads it, Aditi reveals the true value    | Proof they experience themselves, not claims on a slide                                                      | 30 min            |
| 2  | <b>Head-to-head with the lab spectrophotometer:</b> same cuvettes, scatter plot LISA vs lab                       | Turns "cheap gadget" into "credible instrument"                                                              | 1 h               |
| 3  | <b>Sample blank + spike recovery</b>                                                                              | Fixes the paper's known weakness (coloured water). Shows real analytical skill                               | 30 min            |
| 4  | <b>Hindi voice result + WhatsApp share</b> (already in the app)                                                   | Built for the Indian field worker, not just a lab user                                                       | 0 (done)          |
| 5  | <b>"Your water, today":</b> live numbers for Ashoka and Sonipat samples                                           | Local and personal. Judges drink this water                                                                  | 1 h               |
| 6  | <b>JJM field-test-kit reader:</b> the same colour tests, read objectively, with data sent to the reporting system | Adoption channel of 24.8 lakh trained women [9]. FTK results are "indicative" today and need a better reader | Slide             |
| 7  | <b>Community water map</b> with hotspot alerts to panchayats                                                      | Turns individual tests into public-health data (the paper cites GIS integration [1])                         | Mock-up           |
| 8  | <b>Pre-dosed reagent pods:</b> sealed tablets or pouches, no liquid acid for users                                | Safety plus the recurring-revenue business model                                                             | Slide             |
| 9  | <b>Flash-powered v2:</b> light pipe from the phone's LED flash, as in the paper, so no battery                    | Fewer parts, lower cost, fully "phone-native"                                                                | Stretch / roadmap |
| 10 | <b>Schools edition:</b> a flat-pack spectrometer for chemistry classes, like the DBT-Foldscope programme [22]     | A second market plus a science-education story. Foldscope's own path                                         | Slide             |

## The "revolutionary" framing, in one sentence

"Foldscope put a microscope in every pocket for under a dollar [2]. LISA puts a chemistry lab in every pocket: a ₹-cheap spectrometer, a library of chemistry packs, and a map that turns millions of tests into a live picture of India's water."

## 12 Team roles, one card per person

Each person has a clear lane, a clear "done" and a buddy. Aditi and Parth get checklist-style tasks with a named person to ask, so nobody is ever stuck or idle.

### **Yash: Tech lead • app • optics integration • live-demo operator**

- Deploy the app (Section 9.4), test all three modes, and do the simulator walkthrough with the whole team (20 min) so everyone knows the flow.
- With Parth: mount the grating, find the angle, lock the build. Do the wavelength calibration and ROI.
- With Raj: run the calibration and validation (Section 8). Export backups after every milestone.
- Polish list (9.5). Feature freeze at **T+18h**.
- On stage: runs the demo phone; answers app and optics questions.

**Done** = real calibration on the demo phone,  $R^2 \geq 0.99$ , exported JSON, and a 2-minute backup screen recording.

### **Raj: Science lead • chemistry • lab access • safety • reference instrument**

- Book the chemistry lab and UV-Vis spectrophotometer slot. Get the grating from the physics lab and 6+ cuvettes.
- Make reagents and standards (Section 7). Stability test overnight. Prepare 3 blind samples and the spike.
- Read every standard and sample on the lab UV-Vis (650 nm; 880 nm if the instrument allows) for the head-to-head.
- Stretch after Gate 2: the iron pack. Fluoride only if SPADNS is on the shelf.
- On stage: answers chemistry and accuracy questions. Owns the "honest limits" answers (lead, total P vs ortho-P).

**Done** = labelled reagent set, standards, lab reference values in the data sheet, and pre-reacted demo cuvettes.

### **Stuti: Story lead • pitch deck • research • lead presenter**

- Build the deck from Section 15 (design rules included). Get the citations from Section 20 onto the slides.
- Verify every number on a slide against its source. Keep a "claims → source" sheet for Q&A.
- Write and rehearse the story: the opening hook, the problem, the handover to Yash for the demo, the close.
- Coordinate the community-map mock-up and the before/after "FTK colour chart vs LISA" visual.

**Done** = final deck at T+18h (PDF export plus a copy on 2 laptops and a pen drive), a 3-minute and a 5-minute version, and 3 rehearsals.

### **Aditi: Data & proof lead • sample collection • Q&A drill master**

- **Checklist tasks** (buddy: Raj for lab work, Stuti for content):
  1. Set up the data sheet (8.5) in Google Sheets; log every measurement with its time.
  2. Collect and label real samples (7.3) with Parth: ID, place, time, photo.
  3. Label tubes and cuvettes; do water-only dilutions of the standards under Raj's eye (no acids).
  4. Blind-test keeper: write the codes (e.g. X, Y, Z), keep the answer key sealed, and reveal it on stage.
  5. Q&A drill: quiz each teammate from Section 17 during rehearsals and time the answers (under 30 s each).

**Done** = complete data sheet, samples ready, blind kit ready, 2 full Q&A drills run.

### **Parth: Build & logistics lead • enclosure • procurement • video • stage manager**

- **Checklist tasks** (buddy: Yash for the build, Stuti for the video):
  1. Buying run in the first 60 minutes (6.1 list). Keep receipts for the cost slide.
  2. Print Appendix B at 100% scale, check the 50 mm bar, cut, fold and glue enclosure v1, then a cleaner v2.
  3. Film B-roll all day: building, the colour change, the app in use, the team working. This becomes the 60–90 s backup video.
  4. Stage kit from Appendix A: charged phones, power bank, spare LED, cuvettes, tissues, gloves, printed QR.
  5. During the pitch: timekeeper (shows cards: 1 min left, 30 s left) and hands over props.

**Done** = two working enclosures, the backup video exported, the demo kit packed and checked.

## 13 The 24-hour timeline and go/no-go gates

T+0 = kickoff. The amber rows are decision gates. Blue rows are sleep. Every person sleeps at least 3 hours, because a tired presenter loses more points than one extra feature gains.

| T+          | Yash                                                                                                                                                                                                                              | Raj                                                                        | Stuti                                                           | Aditi                                                              | Parth                                              |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------|
| 0:00–0:45   | <b>All:</b> read Sections 1, 2 and 13 • confirm roles • WhatsApp group + shared Drive • simulator walkthrough of the app on Yash's phone • Parth leaves for the buying run with the list                                          |                                                                            |                                                                 |                                                                    |                                                    |
| 0:45–3:00   | Deploy app (Netlify/GitHub); test camera on 2 Android phones                                                                                                                                                                      | Book lab + UV-Vis; get grating and cuvettes; make R1–R3 and P stock        | Deck skeleton, design system, slides 1–3                        | Data sheet; label bottles; start real-sample collection            | Buying run → print template → cut and fold v1      |
| 3:00–5:00   | With Parth: grating on camera, find the angle, fix the wedge, black shroud                                                                                                                                                        | Make standards; react one set; read on lab UV-Vis                          | Problem + market slides (numbers from Sections 1, 18)           | Collect tap / cooler / drain / DAP / detergent samples             | Finish v1 with Yash; start v2                      |
| 5:00        | <b>GATE 1: Is a clean, unsaturated rainbow visible and stable in the app? YES → continue. NO after 45 min of troubleshooting (6.3) → switch to Colorimeter (RGB) mode and continue the same plan.</b>                             |                                                                            |                                                                 |                                                                    |                                                    |
| 5:00–8:00   | CFL wavelength cal; ROI; lock exposure; calibration run (7 std × 3)                                                                                                                                                               | Reacting standards in step with Yash's scans; blind samples X/Y/Z + spike  | Solution / how-it-works slides (Figures 1–2); demo script draft | Logging every point; blind key sealed                              | v2 enclosure (cleaner, painted); B-roll filming    |
| 8:00        | <b>GATE 2: <math>R^2 \geq 0.98</math> and blank repeatability within <math>\pm 0.01</math> A? YES → validation + iron stretch. NO → 1 h of focused fixing (fresh reagents, slit, exposure); still NO → Plan B/C (Section 14).</b> |                                                                            |                                                                 |                                                                    |                                                    |
| 8:00–11:00  | Validation: blind, spike, real samples, repeatability; app polish (timer, logo)                                                                                                                                                   | UV-Vis readings of the same cuvettes; start iron pack if phosphate is done | Results slide with real numbers; competition + business slides  | Enters lab values into History → % error; builds the scatter chart | Films colour change and app use; photos for slides |
| 11:00–14:00 | <b>Sleep</b>                                                                                                                                                                                                                      | <b>Sleep</b> (after starting the stability test)                           | Deck polish; map mock-up                                        | Q&A bank flashcards                                                | Edit backup video (60–90 s)                        |
| 14:00–17:00 | Iron calibration (stretch) or second phosphate run on a fresh blank                                                                                                                                                               | Stability readings (1 h, 3 h); iron standards                              | <b>Sleep</b>                                                    | <b>Sleep</b>                                                       | <b>Sleep</b>                                       |
| 17:00–18:00 | <b>All:</b> review the deck together • lock the numbers on the results slide • <b>FEATURE FREEZE (T+18)</b> . No new code or chemistry after this.                                                                                |                                                                            |                                                                 |                                                                    |                                                    |
| 18:00–20:00 | Record a 2-min screen video of a real measurement (backup)                                                                                                                                                                        | Fresh R3; re-check a standard; prepare demo cuvettes                       | Rehearsal 1 (full run, timed)                                   | Q&A drill 1                                                        | Pack stage kit (App. A)                            |
| 20:00–23:00 | <b>All:</b> Rehearsals 2 and 3 with the live demo • Q&A drill 2 (Aditi + Parth play tough judges) • fix the weak slide • charge everything                                                                                        |                                                                            |                                                                 |                                                                    |                                                    |
| 23:00–24:00 | <b>Buffer.</b> React the demo samples 10–30 min before stage (per the stability test) • quick re-blank on the venue table • breathe                                                                                               |                                                                            |                                                                 |                                                                    |                                                    |

### Communication rhythm

A 5-minute stand-up at every gate and at T+11, T+17 and T+20: *What's done? What's blocked? What's next?* One shared "status" message pinned in WhatsApp. If a task is blocked for more than 30 minutes, escalate to Yash (tech) or Raj (chem) at once.

## 14 Backup plans (fallback ladder)

Go down one level only when a gate fails. Every level is honest and still worth showing.

| Level         | What we show                                                                                             | Trigger                               | What we say                                                                                                   |
|---------------|----------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Plan A</b> | Grating spectrometer + phosphate, live blind test, UV-Vis comparison                                     | —                                     | "Full spectrum, 400–700 nm, calibrated live."                                                                 |
| <b>Plan B</b> | Same app in <b>Colorimeter (RGB) mode</b> : camera looks at the lit cuvette, no grating                  | Gate 1 fails (no clean rainbow)       | "Today's build reads three colour channels; the grating version is next. The same chemistry and model apply." |
| <b>Plan C</b> | Switch the analyte to <b>iron</b> (strongest, most forgiving colour, ~510 nm)                            | Phosphate reagents missing or failing | "Our first pack is iron, an IS 10500 parameter. Phosphate is next."                                           |
| <b>Plan D</b> | Recorded video of a real measurement + real calibration data on slides; live app on a pre-captured photo | Hardware breaks at the venue          | "Here's the real run from last night." Never use the simulator as if it were real.                            |

### Venue risks and fixes

| Risk                                   | Fix                                                                                                                  |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Bright stage lights leak into the tube | Black shroud + blank captured on the stage table right before the demo                                               |
| No Wi-Fi                               | The app works offline once loaded; add it to the home screen in advance. The deck is a PDF on a laptop and pen drive |
| Phone battery / overheating            | 2 charged phones, both calibrated; power bank; screen brightness low                                                 |
| Cuvette spill / breakage               | 6 spare pre-reacted, capped cuvettes; tissues; gloves                                                                |
| Screen mirroring fails                 | Rehearse without mirroring: the judge looks at the phone directly, plus a slide with the backup video                |

# 15 The pitch deck, slide by slide

12 slides for a 5-minute slot (for 3 minutes, drop slides 7, 10 and 11). Owner: Stuti. Build it in Google Slides or Canva.

## Design rules

16:9 • one idea per slide • headline = the takeaway (a full sentence) • at most ~20 words of body text • big numbers • colours: deep teal #0B4F5A, teal #0E7282, amber #E39B00 for highlights, off-white background • one rainbow strip as the brand motif • font: Poppins or Inter (headlines bold), 28pt+ • citation in 10pt grey at the bottom of any slide with a number • real photos from Parth's B-roll beat stock images.

1

Title • 15 s

### LISA: a pocket water lab for every phone

Visual: the device on a phone next to a glass of water; rainbow strip. Team names.

Say: "Would you drink this? It's from our hostel tap. In the next five minutes, you'll know."

2

Problem • 40 s

### India can't see what's in its water

2.1 billion people lack safely managed drinking water [8] • CGWB 2024: nitrate over the limit in 19.8% of samples, iron 13.2%, fluoride 9.04% [7] • **Sonipat is on the fluoride list** [7].

Visual: map of India with the three stats; Sonipat pin.

3

The testing gap • 30 s

### Testing today is either accurate or accessible, never both

Lab: accurate but days away, far, paid per parameter. Field kits: 24.8 lakh women trained [9], but results come from matching colours by eye, are "indicative" and leave no data [9,10]. Visual: an FTK colour chart with a question mark next to a lab machine with a price tag.

4

Insight • 20 s

### Every phone is already 75% of a spectrometer

Light (flash) + detector (camera) + computer. Missing: a ₹-cheap way to split light, plus the right chemistry. Foldscope did this for microscopy for under \$1 [2].

5

Solution • 30 s

### LISA: fold, clip, test

Figure 1 pipeline (simplified) + photo of our build. Chemistry pack → spectrum → AI-calibrated concentration → verdict against the BIS limit → Hindi voice + map.

6

LIVE DEMO • 90 s

### Judge's choice: pick a vial

Hand-off to Yash (Section 16 script). Slide shows only a big "LIVE" and the QR to the app, or the backup video if needed.

7

Proof • 30 s

**₹2,000 kit, lab-level agreement**

Left: our calibration curve ( $R^2$ , LOD). Right: scatter of LISA vs lab UV-Vis with % error. Footnote: the 2026 paper reports under 10% error across phones and devices [1]. **Real numbers only.**

8

Platform • 25 s

**One device, many tests: chemistry packs**

Phosphate ✓ (live) • Iron ✓ or next • Fluoride (Sonipat) → next • Nitrate • Chlorine • Lead (research: pre-concentration). App screenshot of the Tests screen.

9

Why we win • 25 s

**Cheaper than meters, objective unlike colour charts, connected unlike both**

Competition table (Section 18.3), with LISA's column highlighted.

10

Business • 30 s

**Razor and blade: sell the kit, then the reagent packs**

Kit ₹1,500–2,500 • packs of 50 tests • channels: JJM / state water departments (B2G), aquaculture and agri labs, schools, NGOs and RWAs. Unit economics from Section 18. Label estimates as estimates.

11

Impact + roadmap • 20 s

**From one test to a live map of India's water**

Community-map mock-up + timeline: 1 month → 100 real samples vs NABL lab; 3 months → fluoride pack + Sonipat pilot; 12 months → JJM FTK integration pilot.

12

Team + ask • 15 s

**Built in 24 hours by five students. Help us put it in 24 lakh hands.**

Photos + one-line roles. The ask: mentorship, lab-validation partner, pilot village / panchayat intro, seed grant. QR to the app.



## 16 Live demo script (90 seconds)

Rehearse it until it's boring. Yash operates; Stuti narrates; Aditi reveals; Parth hands over props and keeps time.

| Time | Who   | Action / line                                                                                                                                                      |
|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0:00 | Stuti | "This is LISA. Folded card, a razor-blade slit, a ₹50 LED and a grating over the camera." (Parth holds it up.)                                                     |
| 0:10 | Aditi | Offers the tray of 3 capped, coded vials to a judge: "Please pick one. Only I know what's inside." (Pre-reacted per the stability test.)                           |
| 0:20 | Yash  | Inserts the blank → taps ① <b>Capture blank</b> . "First the reference: clean water plus reagent."                                                                 |
| 0:30 | Yash  | Inserts the judge's vial → ② <b>Scan sample</b> . Turns the phone to the judges: "Here's the light that got through. The dip at the red end is the blue compound." |
| 0:45 | Yash  | Reads the result and verdict; taps 🗣️ <b>Hindi</b> , and the phone speaks the result.                                                                              |
| 0:55 | Aditi | Opens the sealed key: "True value: 0.40 mg/L." Pause for reaction.                                                                                                 |
| 1:05 | Stuti | "And here's our hostel tap water, measured this morning" (History tab) "...and Sonipat drain water with a sample blank to cancel its colour."                      |
| 1:20 | Stuti | "Every result can be saved with GPS and shared on WhatsApp. Imagine that from 24 lakh field testers." → next slide.                                                |

### Rules for the demo

- Blank on the venue table 2 minutes before you go on, under the venue's lights.
- Choose blind values inside the calibrated range and above the LOQ (e.g. 0.25 / 0.40 / 0.70 mg P/L).
- If the reading is off by more than 20%: stay calm, re-seat the cuvette, re-scan once. If it's still off: "Real instruments need re-blanking. Here's last night's validation" → slide 7. Honesty beats panic.
- Never argue with a judge. Say "Great point. Here's how we'd handle it..."

### Opening hook options (pick one, deliver it without notes)

1. "Raise your hand if you drank tap water today. Neither you nor I know what was in it. Let's find out."
2. "In 2014, a Stanford lab made a microscope out of paper for under a dollar. We asked: why not a chemistry lab?"
3. "Sonipat, where we're standing, is on the government's list of districts with too much fluoride in the groundwater. How would you test yours today?"

## 17 Judge Q&A bank

Answer in under 30 seconds: direct answer → one piece of evidence → what's next. Assigned answerer in brackets.

### 1. How is this different from a colour-chart field kit? [Stuti]

Same kind of chemistry, but read by a calibrated sensor across the whole spectrum instead of a human eye. That removes the subjective matching error documented in field kits [10]. And every test becomes a data point.

### 2. How accurate is it? [Raj]

Our numbers:  $R^2$  \_\_, LOD \_\_ mg/L, \_\_% average error against the chemistry department's UV-Vis. The 2026 paper achieved under 10% error across phones [1]. Next: 100 real samples against a NABL-accredited lab.

### 3. Phosphate isn't a drinking-water parameter. Why? [Raj]

It's a contamination tracer for sewage, detergent and fertiliser, and it's the chemistry validated in the paper. Iron (an IS 10500 parameter) is our second pack. Fluoride is next.

### 4. Can it detect lead or arsenic? [Raj]

Not today, and we won't claim it. At 10  $\mu\text{g/L}$ , simple colour tests aren't sensitive enough, and even commercial home kits struggle [18]. Our roadmap is pre-concentration or an electrochemical add-on on the same app.

### 5. What does the AI actually do? [Yash]

It learns the calibration from the full spectrum and picks the best model by cross-validation. It also runs automatic quality checks (saturation, misalignment, interference). Next is the paper's CNN approach once we have ~500 spectra.

### 6. Every phone camera is different. [Yash]

Correct, so LISA calibrates per phone with standards (about 30 min, once). The paper showed a cross-device calibration matrix that raises agreement from  $R^2 \approx 0.5$  to  $\approx 0.95$  [1]. That's our shared-model roadmap.

### 7. What about ambient light and auto-exposure? [Yash]

A light-tight black enclosure, exposure lock, and a fresh blank before each sample. Absorbance is a ratio, so slow drift cancels.

### 8. Coloured or muddy water? [Raj]

Filter it, and use a *sample blank* (the same water without reagent) to cancel its own colour. The app flags suspicious absorbance outside the band.

### 9. Are the reagents safe for villagers? [Raj]

The product ships pre-dosed, sealed reagent pods, with no open acid, the same way JJM field kits are already used by trained women. The prototype uses lab reagents handled only by trained people.

### 10. What does it cost at scale? [Stuti]

Prototype ₹\_\_ (receipts). At scale we estimate a few hundred rupees in hardware and a few rupees of chemicals per test. The kit retails at ₹1,500–2,500 with refill packs. These are estimates we'll validate.

### 11. Who pays? [Stuti]

Government water-quality programmes (JJM/state water departments), aquaculture and agriculture labs, schools, NGOs and housing societies. The recurring revenue is reagent packs.

### 12. Why would the government switch from its current kits? [Stuti]

It doesn't have to switch. LISA reads the same kinds of colour tests objectively and digitises them, and FTK results are officially only "indicative" today [9].

### 13. Isn't this just the Chinese paper? [Yash]

The paper proves the physics with a device under \$700. We rebuilt it for a few thousand rupees, Foldscope-style, with India-first chemistry packs, Hindi voice, offline use, the FTK channel and a community map. We cite it proudly.

**14. How long does a test take? [Yash]**

About 5 minutes of reaction time plus seconds to scan. The paper reports under 5 minutes vs over 40 in the lab [1].

**15. Temperature effects? [Raj]**

Colour development depends on temperature and time. The paper found best results above 20°C with a fixed time [1]. We fix the timer and will add temperature compensation in the field version.

**16. Shelf life of the reagents? [Raj]**

Ascorbic acid solution is the weak link, which is why the product uses dry pre-weighed powder or tablets that you add water to.

**17. What's your limit of detection vs the limit that matters? [Raj]**

Ours: \_\_\_ mg/L. The phosphate alert level is 0.1 mg P/L, and iron's is 0.3–1.0 mg/L, so we cover the decision range. For lower limits, use a longer cuvette ( $\ell \uparrow \rightarrow A \uparrow$ ).

**18. IP / defensibility? [Stuti]**

The science is public (like Foldscope's). Our moat is the calibration data per phone, the validated chemistry packs, the distribution partnerships and the data network, which gets better with every test.

**19. How do you know the demo isn't rigged? [Aditi]**

You picked the vial, and the key was sealed before you chose. Please pick another.

**20. Biggest risk? [Yash]**

Field robustness across phones and water matrices. That's why the next milestone is 100 real samples against a NABL lab across 3 phone models.

**21. What's the impact metric? [Stuti]**

Tests run, unsafe sources flagged, sources fixed or switched, and the time from test to action.

**22. Why this team? [Stuti]**

Chemistry lab access and method knowledge (Raj), full-stack and hardware (Yash), research and design (Stuti), with field logistics and data (Aditi, Parth). Built and validated in 24 hours.

# 18 Market, business model, competition

## 18.1 Who needs it

| Segment                                                        | Pain                                                              | What they'd use LISA for                                      |
|----------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------|
| Rural water-quality programmes (JJM / state water departments) | Millions of field tests read by eye; lab confirmation is slow [9] | Objective reading + digital record of field tests             |
| Aquaculture and farms                                          | Phosphate, ammonia and iron affect yields                         | Routine on-farm checks                                        |
| Schools and colleges                                           | Spectrometers are expensive for teaching                          | Flat-pack spectrometer for labs (the Foldscope/DBT path [22]) |
| NGOs, RWAs, households                                         | "Is my borewell or RO water okay?"                                | Quick screening before paying for a lab test                  |

## 18.2 Model and unit economics (estimates, to be validated)

| Item                                                     | Estimate                                  | Basis                                                       |
|----------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------|
| Hardware cost at scale (body, grating, slit, LED, mount) | ₹300–600                                  | Film gratings and die-cut card/plastic in volume            |
| Kit retail                                               | ₹1,500–2,500                              | Well below single-parameter photometers and lab instruments |
| Chemicals per test                                       | ~₹1–5                                     | Milligram-scale reagent use per 10 mL test                  |
| Refill pack (50 tests)                                   | ₹250–500                                  | Recurring revenue ("razor-blade")                           |
| Software                                                 | Free app; paid dashboard for institutions | Maps, alerts, exports for departments / NGOs                |

## 18.3 Competition

|                        | Accredited lab              | Colour-chart field kit      | Single-parameter photometer     | Feng et al. 2026 device [1] | LISA                     |
|------------------------|-----------------------------|-----------------------------|---------------------------------|-----------------------------|--------------------------|
| Accuracy               | Highest                     | Low-medium, subjective [10] | High                            | High (<10% error)           | Medium-high (validating) |
| Time                   | Days                        | Minutes                     | Minutes                         | <5 min                      | ~5–10 min                |
| Cost                   | Per-parameter fees + travel | Cheap                       | A separate device per parameter | <\$700                      | ₹-cheap kit + packs      |
| Multi-test             | Yes                         | Yes                         | No                              | TP only (so far)            | Yes (packs)              |
| Data / map             | Report                      | Manual                      | Some                            | Cloud                       | Built-in, offline-first  |
| Local language / voice | —                           | —                           | —                               | —                           | Hindi voice              |

## 19 Roadmap after the hackathon

| When     | Milestone                                                                    | Proof point                                          |
|----------|------------------------------------------------------------------------------|------------------------------------------------------|
| Month 1  | 100 real samples (Sonipat + campus) vs a NABL-accredited lab; 3 phone models | Error distribution and false-alarm rate              |
| Month 2  | Fluoride pack (SPADNS) + flash-powered v2 optics                             | Sonipat fluoride map, v1                             |
| Month 3  | Pre-dosed reagent pods; cross-phone calibration model                        | A new user gets a correct result without calibrating |
| Month 6  | Pilot with 1–2 panchayats / an NGO; FTK-reader mode                          | Tests per week, sources flagged, actions taken       |
| Month 12 | State-department or JJM pilot; schools edition                               | Institutional purchase order / grant                 |

**Grants and programmes to look at later:** Ashoka's own entrepreneurship programmes, the BIRAC BIG scheme (biotech ignition grants), Atal Innovation Mission, and India-focused water and sanitation challenges. Check current calls before applying.

## 20 References to cite

Put the short form on slides (e.g. "Feng et al., Sens. Actuators B, 2026") and this full list on a final "References" slide or in the handout.

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11. McGonigle AJS, Wilkes TC, Pering TD, et al. Smartphone spectrometers. *Sensors* 18(1) (2018) 223. doi:10.3390/s18010223
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15. Pitula E, Koba M, Smietana M. Which smartphone for a smartphone-based spectrometer? *Optics & Laser Technology* 140 (2021) 107067. doi:10.1016/j.optlastec.2021.107067
16. Das AJ, Wahi A, Kothari I, Raskar R. Ultra-portable, wireless smartphone spectrometer for rapid, non-destructive testing of fruit ripeness. *Scientific Reports* 6 (2016) 32504. doi:10.1038/srep32504
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18. Challenges of detecting lead in drinking water using at-home test kits. *Environmental Science & Technology* (2021). doi:10.1021/acs.est.0c07614
19. Geladi P, Kowalski BR. Partial least-squares regression: a tutorial. *Analytica Chimica Acta* 185 (1986) 1–17.
20. Public Lab. DIY spectrometry kits and Spectral Workbench documentation ([publiclab.org](https://publiclab.org)), for DIY grating and CFL-calibration practice.
21. US EPA. Quality Criteria for Water 1986 ("Gold Book"), phosphorus guidance.
22. Biotech Consortium India Ltd. DBT–Foldscape Programme ([biotech.co.in](https://biotech.co.in)), announced by India's Department of Biotechnology in 2015.

### Citation hygiene (Stuti)

Open each source once before putting its number on a slide. If a number can't be verified, remove it. Judges sometimes check one citation live.

# A Appendix: checklists

## Stage kit (Parth packs it at T+19)

- ☐ Demo phone 1 (calibrated) + phone 2 (calibrated backup)
- ☐ LISA enclosure v1 + v2
- ☐ Power bank + USB LED + spare LED/coin cell
- ☐ 6 capped pre-reacted cuvettes: blank, blind X/Y/Z, tap, drain
- ☐ Sealed blind-answer envelope
- ☐ Tissues, gloves, small waste bag
- ☐ Printed QR to the app + 1-page handout
- ☐ Laptop with deck PDF + backup video; pen drive copy
- ☐ HDMI/USB-C adapters; phone mirroring app tested
- ☐ Time cards (1 min / 30 s)

## Before walking on stage

- ☐ Exposure locked; brightness low; Do Not Disturb on
- ☐ Fresh blank captured on the venue table
- ☐ App open on the Measure screen, test = Phosphate
- ☐ Simulator OFF (badge reads "Camera")
- ☐ Deck on slide 1; backup video queued

## Gate 2 technical checklist

- ☐ Wavelength fit  $R^2 \geq 0.999$
- ☐ Blank peak 150–220, no saturation
- ☐ 7 standards  $\times$  3 replicates captured
- ☐ Calibration  $R^2 \geq 0.98$  (target 0.99)
- ☐ LOD recorded
- ☐ Calibration JSON exported to Drive
- ☐ UV-Vis reference values logged

## Deck final checklist

- ☐ Every number has a source in 10pt
- ☐ Results slide has **our** real data
- ☐ Nothing claims untested analytes work
- ☐ 3-min and 5-min versions timed
- ☐ Team slide + clear ask
- ☐ Exported as PDF; fonts embedded

## Things to write in the blanks on the day

$R^2$  (phosphate)

LOD / LOQ (mg P/L)

Mean % error vs UV-Vis

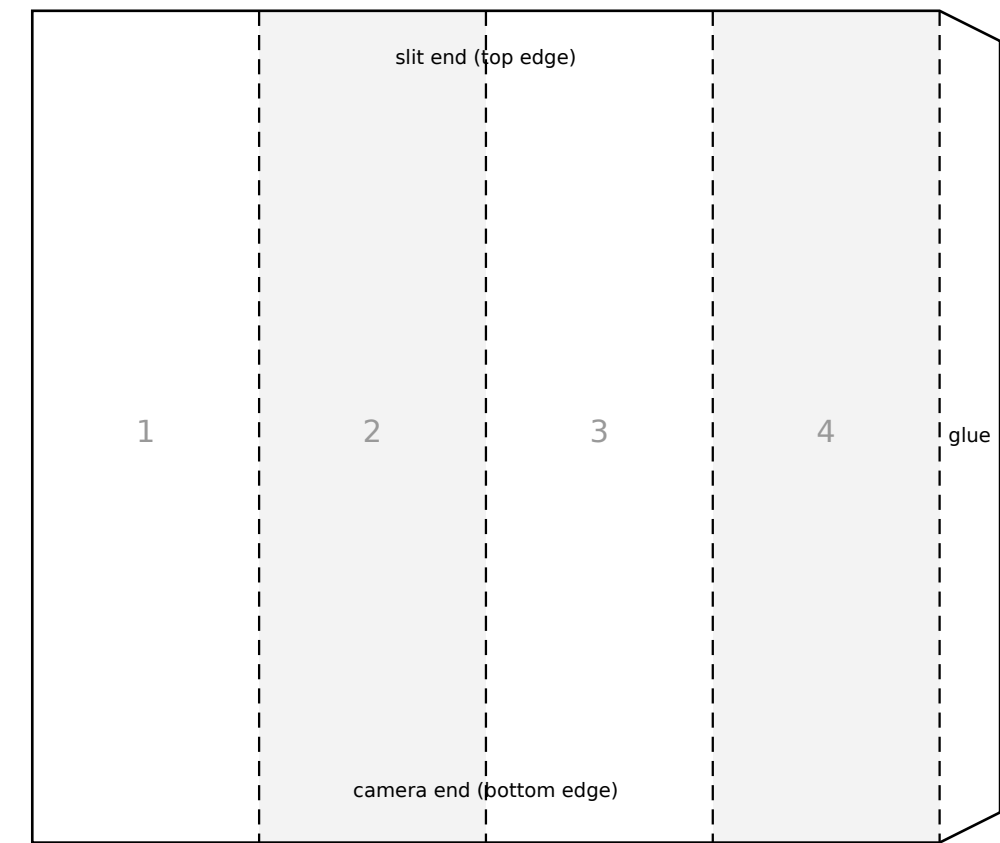
Blind test result / truth

Prototype cost (receipts)

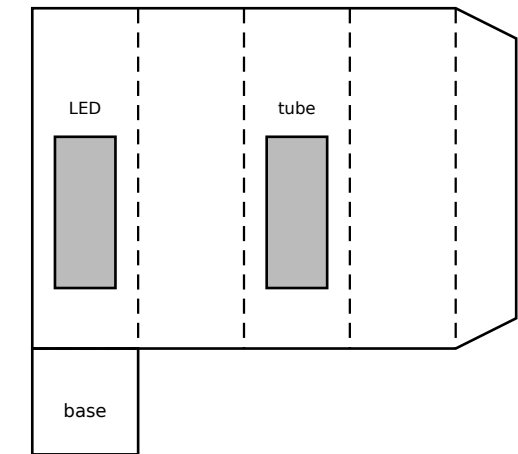
**B • Appendix: printable enclosure template (1:1). Print this page only, at 100% / "Actual size."**

Solid line = cut · dashed line = fold · grey = cut out. Print at 100% / Actual size on A4 (300 gsm card).

**A • Main tube (fold into 30×30 mm square, 110 mm long) — line inside with black tape**



**B • Cuvette chamber**



14×14 mm inside, 45 mm tall; cut out both grey windows

**E • Wedge sides — cut 2 of the angle matching your grating (tube glues on the slope)**

